

ULTRAFILTER LIMITS IN THE PROJECTION LATTICE OF A HILBERT SPACE

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ABSTRACT. A bounded sequence of reals has a limit along any ultrafilter on its index set. We study a linear analogue in which the index set becomes a separable Hilbert space $\mathcal{H}$, subsets become closed subspaces, ultrafilters become maximal filters in the projection lattice $\mathcal{P}(\mathcal{H})$ (*plufs*), bounded sequences become ellipsoids, and convergence becomes the existence of arbitrarily round slices. Three answers emerge. For plufs aligned with an atomic masa, the analogue holds, and is equivalent to the Kadison–Singer problem, resolved by Marcus, Spielman and Srivastava; we record an elementary packaging of that theorem: *every ellipsoid in ℓ^2 has almost-round coordinate slices along every ultrafilter on $\mathbb{N}$* . Under the continuum hypothesis the analogue fails as badly as possible: we construct a pluf π and an ellipsoid E of eccentricity 4 such that every π -slice of E has eccentricity exactly 4; indeed the set of π -limits of the associated quadratic form is an entire interval. Finally, on a Hilbert space of measurable dimension the analogue holds along any normal measure, by a soft diagonal-intersection argument developed in a companion paper. A closing section reads these results as a fragment of “linear model theory”: plufs are maximal consistent incidence types, the state-space face S_π is the space of their completions, Łoś’s theorem holds exactly on the Boolean and normal-measure fragments, and the obstruction to a linear Feferman–Vaught theory is entanglement.

1. INTRODUCTION

Gian-Carlo Rota liked to observe that statements of discrete mathematics often admit formal linear analogues, and that these can turn out to be false, trivial, deep, or merely parallel; even formulating the analogue is not mechanical, despite the functorial passage from a set to the Hilbert space it generates. We take as our discrete statement:

a bounded sequence of reals converges along every ultrafilter on $\mathbb{N}$,

and we linearize every ingredient. Throughout, $\mathcal{H}$ denotes a separable infinite-dimensional real Hilbert space (everything below works verbatim over $\mathbb{C}$), $\mathcal{P}(\mathcal{H})$ its lattice of closed subspaces, and $\mathcal{B}(\mathcal{H})$ the algebra of bounded operators.

Definition 1.1. A *projection-lattice ultrafilter*, or *pluf*, is a family $\pi \subseteq \mathcal{P}(\mathcal{H})$ of nonzero closed subspaces, closed under finite intersection and under enlargement, and maximal with these properties; equivalently, a maximal filter in $\mathcal{P}(\mathcal{H})$.

Plufs exist by Zorn’s lemma. The family $\pi_L = \{M : L \subseteq M\}$ of all closed subspaces containing a fixed line L is a pluf; we call such plufs *principal*.

For the linear analogue of a bounded sequence we take a positive definite bounded symmetric form T with bounded inverse, or equivalently the ellipsoid $E_T = \{x : \langle Tx, x \rangle = 1\}$; diagonal T recovers bounded sequences. Write $m(E)$ and $M(E)$ for the minor and major radii of an ellipsoid and $r(E) = M(E)/m(E)$ for its eccentricity. Slicing by a closed subspace never increases eccentricity.

Definition 1.2. A pluf π has the *round slice property* (RSP) for T if $\inf_{M \in \pi} r(E_T \cap M) = 1$.

The proposed analogue is: *every pluf has RSP for every T* . This paper establishes the following trichotomy.

- (A) (§4) Every pluf containing the coordinate subspaces of an ultrafilter on a fixed orthonormal basis has RSP for every T . This is exactly the Kadison–Singer problem [10], resolved by Marcus, Spielman and Srivastava [14]; conversely, the RSP statement for coordinate filters is equivalent to Kadison–Singer (Remark 4.2).
- (B) (§5) Assuming the continuum hypothesis, there is a pluf π and an ellipsoid E of eccentricity 4 such that *every* π -slice of E has minor radius 1, major radius 4, and eccentricity 4. The failure is total: the set of π -limits of the form is the entire interval between the essential extremes (Proposition 5.11).
- (C) (§6) If $\dim \mathcal{H}$ is a measurable cardinal and $\mathcal{U}$ is a normal measure on it, the coordinate filter of $\mathcal{U}$ is *already* a pluf, and it has RSP for every T , by a diagonal-intersection argument requiring no paving theorem. Full proofs appear in the companion paper [9].

The reading in §7 is model-theoretic: a pluf is a maximal consistent set of *incidence* conditions, the face S_π of states fixing it (§3) is the space of completions of that type, and RSP is Łoś’s theorem for atomic formulas. Łoś survives linearization exactly on the Boolean fragment (by Marcus–Spielman–Srivastava) and along normal measures (softly); in between, the gap is measured by an ellipsoid.

A companion paper [8] develops the combinatorial theory that determines *which* plufs arise from ultrafilters on bases, characterizes maximality of the block-generated filter by selectivity of the ultrafilter, and reduces the diagonalizability of a pluf to the existence of a single *intimate* witness. The nearest published relative of our maximal lattice filters is Question 6.39 of Farah–Wofsey [6], which asks whether every maximal filter in the projection lattice of the *Calkin* algebra lifts; we discuss the bridge in §8.

2. STRUCTURE OF THE SPACE OF PLUFS

The facts below are elementary. They already show why the Stone–Čech analogy breaks, before any analysis enters.

Lemma 2.1. *A filter π in $\mathcal{P}(\mathcal{H})$ is maximal if and only if for every $M \in \mathcal{P}(\mathcal{H})$ either $M \in \pi$ or there is $N \in \pi$ with $M \cap N = 0$.*

Proof. The filter generated by $\pi \cup \{M\}$ consists of the subspaces containing some $M \cap N$ with $N \in \pi$, and is proper exactly when no such intersection vanishes. $\square$

The condition $M \cap N = 0$ is far weaker than $N \subseteq M^\perp$: two subspaces can meet trivially while lying at angle 0. The lattice sees incidence and forgets angle.

Lemma 2.2. *If a pluf π contains a finite-dimensional subspace then $\pi = \pi_L$ for a line L . Consequently a pluf is principal if and only if $\bigcap \pi \neq 0$, in which case $\bigcap \pi$ is a line; and every nonprincipal pluf contains every subspace of finite codimension.*

Proof. Choose $M \in \pi$ of least finite dimension d . For $N \in \pi$, $M \cap N \in \pi$ has dimension at most d , hence exactly d , hence $M \subseteq N$. If $d \geq 2$, any line $L \subsetneq M$ meets every member of π , contradicting maximality; so $d = 1$ and $\pi = \pi_M$. If $0 \neq v \in \bigcap \pi$, maximality puts $\mathbb{R}v$ into π . For the last claim: if V has finite codimension and $V \notin \pi$, some $M \in \pi$ satisfies $M \cap V = 0$, forcing $\dim M < \infty$ and principality. $\square$

Proposition 2.3. $\mathcal{P}(\mathcal{H})$ has no prime filters: no proper filter F satisfies $M \vee N \in F \Rightarrow M \in F$ or $N \in F$. The same holds for the lattice of closed subspaces of any real Hilbert space with an infinite orthonormal basis, and for the subspace lattice of any real inner product space of finite dimension at least two.

Proof. Suppose F is prime. Since F is nonempty and upward closed, the whole space lies in F . In the presence of an infinite orthonormal basis, fix three closed subspaces A, B, C that pairwise meet trivially and pairwise span: indexing the basis as (e_n) , take A the closed span of the even-indexed vectors, B that of the odd-indexed ones, and

$$C = \{x : x_{2n} = x_{2n+1} \text{ for all } n\},$$

so that $A \cap B = A \cap C = B \cap C = 0$ while $A \vee B = A \vee C = B \vee C = \mathcal{H}$ (for the last two, $e_{2n+1} = (e_{2n} + e_{2n+1}) - e_{2n}$ and symmetrically). Applying primeness to the three joins puts one member of each pair $\{A, B\}$, $\{A, C\}$, $\{B, C\}$ into F ; whichever way the choices fall, F contains two of A, B, C , whose intersection is 0 — contradicting that F is a proper filter.

A triple as above requires $\dim A = \dim B = \dim C$, so none exists in odd finite dimension. There the following argument serves, and covers every finite dimension at least two. A prime filter of a finite-dimensional lattice contains a finite-dimensional member, hence is principal at a line $\mathbb{R}v$ by Lemma 2.2. Choose $w \notin \mathbb{R}v$, put $S = \text{span}\{v, w\}$ and let C be a complement of S . Then

$$P = \mathbb{R}w \oplus C, \quad Q = \mathbb{R}(v + w) \oplus C$$

satisfy $P \vee Q \supseteq \mathbb{R}w + \mathbb{R}(v + w) + C = S + C = \mathcal{H}$, while $v \notin P$ and $v \notin Q$: a relation $v = aw + c$ with $c \in C$ forces $c = v - aw \in S \cap C = 0$ and hence $w \in \mathbb{R}v$, and symmetrically for Q . Primeness applied to $P \vee Q$ would put $\mathbb{R}v$ inside P or inside Q , which it is not.

Primeness of a filter through all of $\mathcal{P}(\mathcal{H})$ would furnish a dispersion-free $\{0, 1\}$ -valued lattice homomorphism, forbidden for $\dim \mathcal{H} \geq 3$ by Kochen–Specker [11]. The arguments above are elementary substitutes covering also $\dim \mathcal{H} = 2$. $\square$

Topologize the set Π of plufs by the basis $\widehat{M} = \{\pi : M \in \pi\}$; note $\widehat{M} \cap \widehat{N} = \widehat{M \cap N}$.

Proposition 2.4. Π is Hausdorff and zero-dimensional; the principal plufs form a dense set of isolated points of cardinality $2^{\aleph_0}$; and Π is not compact.

Proof. Each $\widehat{M}$ is clopen, with complement $\bigcup\{\widehat{N} : N \cap M = 0\}$ by Lemma 2.1; two distinct plufs are separated by an $\widehat{M}, \widehat{N}$ with $M \cap N = 0$. For a line L , $\widehat{L} = \{\pi_L\}$, so principal plufs are isolated, and they are dense since $\pi_L \in \widehat{M}$ for every line $L \subseteq M$. If $\dim M = 2$ then by Lemma 2.2 $\widehat{M}$ consists exactly of the π_L , $L \subseteq M$: an uncountable discrete closed subspace, so Π is not compact. $\square$

Thus Π contains the projective space $\mathbf{P}(\mathcal{H})$ densely and *discretely* — its own topology destroyed — with no compactification structure. Since ultrafilter limits are nothing but the compactness of $\beta\mathbb{N}$, the RSP cannot be had softly.

3. SLICES, GAPS, AND STATES

Fix a positive $T \in \mathcal{B}(\mathcal{H})$ with bounded inverse and set, for $0 \neq M \in \mathcal{P}(\mathcal{H})$,

$$u_T(M) = \sup_{x \in M, \|x\|=1} \langle Tx, x \rangle, \quad \ell_T(M) = \inf_{x \in M, \|x\|=1} \langle Tx, x \rangle,$$

so that with $T_M = P_M T P_M|_M$ we have $u_T(M) = \max \sigma(T_M)$, $\ell_T(M) = \min \sigma(T_M)$.

Lemma 3.1. $m(E_T \cap M) = u_T(M)^{-1/2}$, $M(E_T \cap M) = \ell_T(M)^{-1/2}$, and $r(E_T \cap M) = \sqrt{u_T(M)/\ell_T(M)}$.

Downward directedness of a filter gives $\sup_{M \in \pi} \ell_T(M) \leq \inf_{M \in \pi} u_T(M)$, whence:

Proposition 3.2. π has RSP for T iff $\sup_{M \in \pi} \ell_T(M) = \inf_{M \in \pi} u_T(M)$, iff for every $\epsilon > 0$ some $M \in \pi$ has $u_T(M) - \ell_T(M) < \epsilon$.

The simultaneous statement over all T is state-theoretic. Put

$$S_\pi = \{\varphi \text{ a state on } \mathcal{B}(\mathcal{H}) : \varphi(P_M) = 1 \text{ for all } M \in \pi\}.$$

Proposition 3.3. S_π is a nonempty weak*-compact face of the state space of $\mathcal{B}(\mathcal{H})$; it equals the set of states with $\sup_M \ell_T(M) \leq \varphi(T) \leq \inf_M u_T(M)$ for every self-adjoint T ; and π has RSP for every T iff S_π is a singleton, in which case its element is a pure state.

Proof. $\overline{\lim}_\pi T := \inf_M u_T(M)$ is sublinear on self-adjoints, with $\underline{\lim}_\pi T = -\overline{\lim}_\pi(-T)$; Hahn–Banach supplies a linear functional between them, which is a state, so $S_\pi \neq \emptyset$. If $\varphi(P_M) = 1$ then $\varphi(A) = \varphi(P_M A P_M)$ and $\ell_T(M)P_M \leq P_M T P_M \leq u_T(M)P_M$ gives the sandwich; conversely $\ell_{P_M}(M) = 1$ forces $\varphi(P_M) = 1$. A convex combination is 1 at P_M only if both summands are, so S_π is a face and its extreme points are pure. The last sentence is Proposition 3.2 over all T . $\square$

The face S_π realizes the pluf: in the GNS representation $(\mathcal{H}_\varphi, \rho_\varphi, \xi_\varphi)$ of any $\varphi \in S_\pi$ one has $\rho_\varphi(P_M)\xi_\varphi = \xi_\varphi$ for all $M \in \pi$, i.e. the GNS space contains a vector lying in every subspace of the pluf. For principal $\pi_{\mathbb{R}v}$ this returns $\mathcal{H}$ with $\xi = v$. We return to this reading in §7.

Finally, the pluf notion sits properly inside Farah and Weaver’s *quantum filters* [6, 16]: upward-closed families of projections with $\|p_1 \cdots p_n\| = 1$ for all finite subfamilies, whose maximal elements correspond bijectively to pure states. Every filter in $\mathcal{P}(\mathcal{H})$ is a quantum filter; the converse fails because $\|pq\| = 1$ sees angle where $p \wedge q \neq 0$ sees incidence, and quantum filters need not be closed under meets.

Remark 3.4 (Round subspaces are abundant). Definition 1.2 is not Dvoretzky’s theorem in disguise: for ellipsoids the corresponding fact is stronger and elementary. If Q has eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$, choose $c \in [\lambda_{n+1-k}, \lambda_k]$ and in each plane $\text{span}(e_j, e_{n+1-j})$, $j \leq k \leq n/2$, a unit f_j with $Q(f_j) = c$: the f_j are Q -orthogonal, and Q restricts to $c \cdot I$ on their span — an *exactly* spherical section of dimension $\lfloor n/2 \rfloor$. The same pairing in $\mathcal{H}$ applied to $T = \text{diag}(1, \frac{1}{16}, 1, \frac{1}{16}, \dots)$ produces an infinite-dimensional subspace on which T compresses to $\frac{1}{2}I$. What §5 constructs is thus a maximal filter dodging a $2^{\aleph_0}$ -sized family of perfectly round subspaces.

4. THE ATOMIC MASA: KADISON–SINGER AS A ROUND-SLICE THEOREM

Fix an orthonormal basis (e_n) of $\mathcal{H}$ and for $S \subseteq \mathbb{N}$ let $\mathcal{H}_S = \overline{\text{span}}\{e_n : n \in S\}$. An ultrafilter $\mathcal{U}$ on $\mathbb{N}$ generates the filter $\{\mathcal{H}_S : S \in \mathcal{U}\}$ (note $\mathcal{H}_S \cap \mathcal{H}_{S'} = \mathcal{H}_{S \cap S'}$).

Theorem 4.1. Let $\mathcal{U}$ be an ultrafilter on $\mathbb{N}$ and π any pluf containing $\{\mathcal{H}_S : S \in \mathcal{U}\}$. Then π has RSP for every T , and $\lim_\pi T = \lim_{n \rightarrow \mathcal{U}} \langle T e_n, e_n \rangle$.

Proof. By Proposition 3.3, S_π lies inside the set of state extensions to $\mathcal{B}(\mathcal{H})$ of the pure state $\mathcal{U}$ on the atomic masa. By the Kadison–Singer problem [10], resolved affirmatively by Marcus, Spielman and Srivastava [14] via Anderson’s paving reformulation [2], that set is a singleton; being nonempty, S_π is that singleton, namely $\mathcal{U} \circ \mathcal{E}$ for the diagonal conditional expectation $\mathcal{E}$. $\square$

Remark 4.2. Unwinding, RSP along coordinate filters restates paving: Anderson's theorem partitions $\mathbb{N}$ into finitely many pieces on each of which $T - \mathcal{E}(T)$ compresses to small norm; one piece lies in $\mathcal{U}$, and shrinking further makes the diagonal nearly constant. Thus the statement

for every ultrafilter $\mathcal{U}$ on $\mathbb{N}$, every ellipsoid E in ℓ^2 , and every $\epsilon > 0$ there is $S \in \mathcal{U}$ with

$$r(E \cap \mathcal{H}_S) < 1 + \epsilon$$

is *equivalent* to the Kadison–Singer problem.

Which plufs contain a coordinate filter, for which the block-generated filter is already maximal, and whether *every* pluf is so aligned, are the subject of the companion paper [8]: the block filter of $\mathcal{U}$ (blocks together with all finite-codimension subspaces) is a pluf iff $\mathcal{U}$ is selective, and a pluf is diagonalizable over a basis iff every member is *intimate* with respect to it.

5. A PLUF WITH NO ROUND SLICES, UNDER CH

Fix once and for all

$$T = \text{diag}\left(1, \frac{1}{16}, 1, \frac{1}{16}, \dots\right), \quad E = E_T,$$

so $\sigma(T) = \sigma_{\text{ess}}(T) = \{\frac{1}{16}, 1\}$, $m(E) = 1$, $M(E) = 4$, $r(E) = 4$.

Definition 5.1. A closed subspace $M \neq 0$ is *ample* if $\{1, \frac{1}{16}\} \subseteq \sigma_{\text{ess}}(T_M)$.

Since $\frac{1}{16} \leq T \leq 1$, ampleness forces $u_T(M) = 1$, $\ell_T(M) = \frac{1}{16}$, so by Lemma 3.1:

Lemma 5.2. *Every ample M satisfies $m(E \cap M) = 1$, $M(E \cap M) = 4$, $r(E \cap M) = 4$; in particular ample subspaces are infinite-dimensional.*

Lemma 5.3 (Upward inheritance). *Let $W \subseteq V$ be closed and $\lambda \in \{\frac{1}{16}, 1\}$. If $\lambda \in \sigma_{\text{ess}}(T_W)$ then $\lambda \in \sigma_{\text{ess}}(T_V)$. Hence ampleness passes upward and its failure passes downward.*

Proof. Since $\sigma(T) = \{\frac{1}{16}, 1\}$ we have $(T - 1)(T - \frac{1}{16}) = 0$, whence for every unit x the two exact identities

$$\|Tx - x\|^2 = \frac{15}{16}(1 - \langle Tx, x \rangle), \quad \|Tx - \frac{1}{16}x\|^2 = \frac{15}{16}(\langle Tx, x \rangle - \frac{1}{16}).$$

Choose a weakly null orthonormal (x_k) in W with $\langle Tx_k, x_k \rangle \rightarrow \lambda$. The identity at λ gives $\|Tx_k - \lambda x_k\| \rightarrow 0$, hence $\|T_V x_k - \lambda x_k\| = \|P_V(Tx_k - \lambda x_k)\| \rightarrow 0$: a weakly null approximate eigensequence for T_V at λ . $\square$

Lemma 5.4. *If $W \subseteq V$ has finite codimension in V then $\sigma_{\text{ess}}(T_W) = \sigma_{\text{ess}}(T_V)$; in particular ampleness is inherited by finite-codimension subspaces.*

Proof. $P_V - P_W$ has finite rank, so T_V and $T_W \oplus 0$ differ by a finite-rank operator. $\square$

Lemma 5.5 (Escape). *Let V be ample, N closed, $V \cap N$ not ample, and $\lambda \in \{\frac{1}{16}, 1\}$ with $\lambda \notin \sigma_{\text{ess}}(T_{V \cap N})$. Then for every $\epsilon > 0$ the spectral subspace $K_\lambda^\epsilon(V) = \mathbf{1}_{[\lambda - \epsilon, \lambda + \epsilon]}(T_V)V$ is infinite-dimensional, $\lambda \in \sigma_{\text{ess}}(T_{K_\lambda^\epsilon(V)})$, and $K_\lambda^\epsilon(V) \cap N$ has infinite codimension in $K_\lambda^\epsilon(V)$. In particular N has infinite codimension in $\mathcal{H}$.*

Proof. Take $\lambda = 1$. A weakly null orthonormal (x_k) in V with $\langle T_V x_k, x_k \rangle \rightarrow 1$ satisfies $\|T_V x_k - x_k\| \rightarrow 0$ by the estimate above, hence $\|\mathbf{1}_{[1 - \epsilon, 1]}(T_V)x_k - x_k\| \rightarrow 0$; so $K := K_1^\epsilon(V)$ is infinite-dimensional with $1 \in \sigma_{\text{ess}}(T_K)$. If $K \cap N$ had finite codimension in K , Lemmas 5.4 and 5.3 would give $1 \in \sigma_{\text{ess}}(T_{V \cap N})$, contrary to hypothesis. $\square$

Remark 5.6. The proof uses only three properties of K : infinite dimension, the estimate $\|Tx - \lambda x\| \leq \delta \|x\|$ for all $x \in K$ (with δ at the caller's disposal), and infinite codimension of $K \cap N$ in K . A subspace with these properties is available without the Borel functional calculus. Since $\lambda \in \sigma_{\text{ess}}(T_V)$, the identities in the proof of Lemma 5.3 supply an orthonormal (u_n) in V with $\|Tu_n - \lambda u_n\| < \delta_n$ for any prescribed $\delta_n > 0$; taking $\sum_n \delta_n \leq \delta$ and K the closed span of the u_n , every $x = \sum_n c_n u_n \in K$ has $|c_n| \leq \|x\|$ and hence $\|Tx - \lambda x\| \leq \sum_n |c_n| \delta_n \leq \delta \|x\|$.

Lemma 5.7 (Blocking lemma). *Let G be a countable, downward directed family of ample subspaces and N a closed subspace such that $q \cap N$ is not ample for some $q \in G$ while $g \cap N \neq 0$ for all $g \in G$. Then there is a closed subspace R with*

- (i) $R \cap N = 0$;
- (ii) $R \cap g$ is ample for every $g \in G$, and R is ample.

Proof. Enumerate $G = \{g_1, g_2, \dots\}$ and put $h_n = q \cap g_1 \cap \dots \cap g_n$: a decreasing sequence in G (directedness), each ample, each $h_n \cap N$ not ample (Lemma 5.3). Fix $\lambda_0 \in \{\frac{1}{16}, 1\}$ with $\lambda_0 \notin \sigma_{\text{ess}}(T_{q \cap N})$, hence $\lambda_0 \notin \sigma_{\text{ess}}(T_{h_n \cap N})$ for all n , and let λ_1 be the other value. Choose $\epsilon_n \downarrow 0$.

We construct unit vectors $w_n \in h_n$ recursively so that, writing $y_n = P_{N^\perp} w_n$:

- (a) $|\langle Tw_n, w_n \rangle - \mu_n| < \epsilon_n$, where $\mu_n = \lambda_0$ for even n and $\mu_n = \lambda_1$ for odd n ;
- (b) $\langle w_n, w_j \rangle = 0$ and $\langle Tw_n, w_j \rangle = 0$ for all $j < n$;
- (c) $y_n \neq 0$ and $\langle w_n, y_j \rangle = 0$ for all $j < n$.

All conditions in (b) and (c) other than $y_n \neq 0$ are *finitely many linear constraints* on w_n (self-adjointness of T converts $\langle Tw_n, w_j \rangle = 0$ into $w_n \perp Tw_j$), so they carve out a finite-codimension subspace C_n of any infinite-dimensional subspace we work in.

Even n . Work in $K := K_{\lambda_0}^{\epsilon_n}(h_n)$, infinite-dimensional with $K \cap N$ of infinite codimension in K (Lemma 5.5). Then $C_n \subseteq K$ is infinite-dimensional and $C_n \cap N \subseteq K \cap N$ has infinite codimension in C_n ; pick a unit $w_n \in C_n \setminus N$, so $y_n \neq 0$; and every unit vector of K satisfies (a) since $\lambda_0 - \epsilon_n \leq T_{h_n} \leq \lambda_0 + \epsilon_n$ on K gives $|\langle Tw_n, w_n \rangle - \lambda_0| \leq \epsilon_n$.

Odd n . First pick a unit $w'_n \in K_{\lambda_1}^{\epsilon_n/2}(h_n)$ satisfying the linear constraints of (b)–(c); then pick a unit $z_n \in K_{\lambda_0}^{\epsilon_n/2}(h_n) \setminus N$ satisfying the same linear constraints together with $z_n \perp w'_n$ and $z_n \perp Tw'_n$ (again finitely many linear conditions inside an infinite-dimensional escaping subspace, by Lemma 5.5). Set $w_n = (w'_n + \eta z_n) / \|w'_n + \eta z_n\|$ for small $\eta > 0$. Conditions (b)–(c) hold for every η by linearity; $y_n \neq 0$ for all but at most one value of η (as $P_{N^\perp} z_n \neq 0$); and $\langle Tw_n, w_n \rangle = (\langle Tw'_n, w'_n \rangle + \eta^2 \langle Tz_n, z_n \rangle) / (1 + \eta^2)$ lies within ϵ_n of λ_1 for η small. This perturbation step replaces the orthogonal decomposition of h_n along N , which non-distributivity forbids.

Let $R = \overline{\text{span}}\{w_n\}$ and $R_j = \overline{\text{span}}\{w_n : n \geq j\}$.

(i). By (b) the w_n are orthonormal, so each $x \in R$ has a unique expansion $x = \sum c_n w_n$ with $(c_n) \in \ell^2$, and $P_{N^\perp} x = \sum c_n y_n$ (the series converges since $\|y_n\| \leq 1$). By (c) the y_n are pairwise orthogonal ($\langle y_n, y_j \rangle = \langle w_n, P_{N^\perp} y_j \rangle = \langle w_n, y_j \rangle = 0$ for $j < n$, as $y_j \in N^\perp$), and each $y_n \neq 0$; so $P_{N^\perp} x = 0$ forces $c_n \|y_n\| = 0$, i.e. $c_n = 0$, for all n . Thus $R \cap N = 0$.

(ii). By (b), T_R is *exactly diagonal* in the orthonormal basis (w_n) of R : $\langle T_R w_m, w_n \rangle = \langle Tw_m, w_n \rangle = 0$ for $m \neq n$ (using self-adjointness for $m < n$), with diagonal entries $d_n = \langle Tw_n, w_n \rangle$ accumulating at both λ_0 and λ_1 by (a). Hence $\sigma_{\text{ess}}(T_{R_j}) \supseteq \{\frac{1}{16}, 1\}$ for every j : R_j is ample. Since $w_n \in h_n \subseteq g_j$ for $n \geq j$, $R_j \subseteq R \cap g_j$, and Lemma 5.3 makes $R \cap g_j$ ample; $j = 1$ gives ampleness of R . $\square$

Remark 5.8. Every constraint imposed in the recursion is linear, so exact orthogonality costs nothing and no error accumulates: T_R is exactly diagonal in the basis (w_n) . Part (ii) consumes only the diagonal values d_n and their accumulation at $\frac{1}{16}$ and 1.

Theorem 5.9. *Assume CH. There is a nonprincipal pluf π every member of which is ample. Consequently every π -slice of E has minor radius 1, major radius 4, and eccentricity 4; in particular π fails RSP for T .*

Proof. Under CH enumerate $\mathcal{P}(\mathcal{H}) = \{p_i : i < \omega_1\}$. We build an increasing chain $(F_i)_{i < \omega_1}$ of countable, intersection-closed families of ample subspaces with:

(*) for each i , either $p_i \in F_i$ or some $g \in F_i$ has $g \cap p_i = 0$.

Set $F_0 = \{\mathcal{H}\}$ and take unions at limits (countable by CH). Given $G = \bigcup_{j < i} F_j$:

Case I: $g \cap p_i = 0$ for some $g \in G$. Set $F_i = G$.

Case II: $g \cap p_i$ is ample for every $g \in G$. Set $F_i = G \cup \{p_i\} \cup \{g \cap p_i : g \in G\}$ (note $p_i = \mathcal{H} \cap p_i$ is ample).

Case III: otherwise. All $g \cap p_i \neq 0$ but some $q \cap p_i$ is not ample. Apply Lemma 5.7 with $N = p_i$ to get R , and set $F_i = G \cup \{R\} \cup \{g \cap R : g \in G\}$.

In each case F_i is countable, intersection-closed, ample, and satisfies (*). Let π be the upward closure of $\bigcup_i F_i$: a proper filter of ample (hence nonzero) subspaces, every member of which is ample by Lemma 5.3, maximal by (*) and Lemma 2.1, and nonprincipal since ample subspaces are infinite-dimensional (Lemma 2.2). Lemma 5.2 gives the radius assertions. $\square$

Remark 5.10. The pluf of Theorem 5.9 contains no coordinate subspace $\mathcal{H}_S$ with S almost contained in the evens or the odds, so its trace on the atomic masa is a filter on $\mathbb{N}$ that is *not* an ultrafilter: the blocking in Case III is performed by subspaces transverse to the masa. Lattice maximality never forces a pluf to decide S against $\mathbb{N} \setminus S$; that is why Theorem 4.1 does not apply.

The failure is not merely the absence of a limit; the “limit” is an entire interval.

Proposition 5.11. *For the pluf π of Theorem 5.9, $\{\varphi(T) : \varphi \in S_\pi\} = [\frac{1}{16}, 1]$: every value between the essential extremes is a π -limit of T .*

Proof. Containment $\subseteq$ is Proposition 3.3 with $u_T \equiv 1$, $\ell_T \equiv \frac{1}{16}$ on π . For $\supseteq$, fix $c \in [\frac{1}{16}, 1]$ and define $\psi(bT) = bc$ on the line $\mathbb{R}T$. For $b \geq 0$, $\overline{\lim}_\pi(bT) = b \geq bc$; for $b < 0$, $\overline{\lim}_\pi(bT) = \frac{b}{16} \geq bc$. So ψ is dominated by the sublinear functional $\overline{\lim}_\pi$ and extends by Hahn–Banach to a linear φ with $\underline{\lim}_\pi \leq \varphi \leq \overline{\lim}_\pi$ on self-adjoints; such φ is a state in S_π (Proposition 3.3) with $\varphi(T) = c$. $\square$

6. MEASURABLE DIMENSION: A SUMMARY

The third act takes place on $\mathcal{H} = \ell^2(\kappa)$ with κ a measurable cardinal and $\mathcal{U}$ a normal measure on κ ; full proofs appear in the companion note [9], whose starting point is the following observation. Every vector of $\ell^2(\kappa)$ has countable support, and a normal measure is closed under diagonal intersections; together these give, for any $T \in \mathcal{B}(\ell^2(\kappa))$, a set $S \in \mathcal{U}$ with $P_S T P_S$ *exactly diagonal* — Anderson paving with error zero, by a diagonal intersection. From this one deduces an exact dichotomy: for every closed subspace W there is $S \in \mathcal{U}$ with either $\mathcal{H}_S \subseteq W$ or $W \cap \mathcal{H}_S = 0$. Consequently:

Theorem 6.1 (see [9]). *Let κ be measurable and $\mathcal{U}$ normal on κ . Then $\Phi(\mathcal{U}) = \{M : \mathcal{H}_S \subseteq M \text{ for some } S \in \mathcal{U}\}$ is a nonprincipal, κ -complete pluf; it has RSP for every T , with $\lim_{\Phi(\mathcal{U})} T = \lim_{\mathcal{U}} \langle T e_\alpha, e_\alpha \rangle$; and $S_{\Phi(\mathcal{U})}$ is a singleton whose element is a singular, countably additive pure state.*

Three qualifications bound the scope of Theorem 6.1. The collapse is local to the normal measure: pure states on the diagonal of $\mathcal{B}(\ell^2(\kappa))$ corresponding to arbitrary ultrafilters on κ (for

instance, those concentrating on a countable set) still require Marcus–Spielman–Srivastava for unique extension, as in Blecher–Weaver [5], Theorem 5.1. Nor is normality idle: maximality of the block filter *forces* a Q-point-type regularity on the measure (every countable-to-one function on κ is injective on a measure-one set), so that maximality is pinched between this property and normality; see [9] for the proof and for the open sufficiency question. The pluf space over $\ell^2(\kappa)$ remains non-compact and prime-filter-free (the arguments of §2 are dimension-free). And non-diagonalizable plufs are not excluded at κ : graphs of unitaries between complementary blocks are non-intimate there just as at $\aleph_0$ [8].

The state $\varphi_U = \lim_U \langle \cdot e_\alpha, e_\alpha \rangle$ is countably additive by κ -completeness and singular because every projection of countable rank is annihilated; it is thus exactly the boundary object of Gleason’s theorem, which forces countably additive states on $\mathcal{B}(\mathcal{H})$ to be normal precisely when $\dim \mathcal{H}$ is non-measurable. The second author of [5] together with Farah developed the quantum-filter side of this boundary; the lattice side, developed in [9], is that at measurable κ — and, below the first Ulam measurable cardinal, *only* there — quantum-filter maximality and lattice maximality coincide, the bridge being the excision theory of [5].

7. A MODEL-THEORETIC CODA

The results above admit a model-theoretic reading. It requires no new theorems, only a dictionary.

Continuous model theory [3, 7] linearizes *structures*: metric domains, uniformly continuous function symbols (our operators $\mathcal{H}^{\otimes n} \rightarrow \mathcal{H}$), $[0, 1]$ -valued predicates, sup / inf quantifiers. Kuperberg–Weaver’s quantum relations [13] linearize *relations*: operator bimodules, reducing over the atomic masa to ordinary relations on $\mathbb{N}$. Both keep the *index* of an ultraproduct classical. The pluf proposal linearizes the ultrafilter itself, and the present paper computes what happens.

In the dictionary: a pluf is a maximal consistent set of incidence conditions “ $v \in M$ ” — a maximal quantifier-free partial type in the incidence fragment; S_π is the space of its completions to full types over $\mathcal{B}(\mathcal{H})$ (pure states being the complete types); the GNS construction is the linear ultrapower, realizing the type (principal plufs give realized types, and the GNS vector ξ_φ of §3 is the realizing element); and RSP is Łoś’s theorem for atomic formulas. Then: Proposition 2.3 says Łoś fails for disjunction — unavoidably, by Kochen–Specker, so any honest linear semantics is lattice-valued, as in Takeuti–Ozawa quantum set theory [15]; Proposition 2.4 says the compactness theorem fails for the fully linearized logic, quantum filters being the metric repair that buys it back; Theorem 4.1 says the Boolean reduct of a type determines its completion (Łoś recovered on the Boolean fragment, at the cost of [14]); Theorem 5.9 says incidence-maximality is consistently far from semantic completeness, with the gap an interval (Proposition 5.11); and Theorem 6.1 says Łoś holds outright along normal measures.

A further obstruction appears at the next step. Products of structures require product plufs on $\mathcal{H} \otimes \mathcal{H}$; but the filter generated by $\{\overline{M \otimes N} : M \in \pi_1, N \in \pi_2\}$ never decides an entangled subspace, and almost all of $\mathcal{P}(\mathcal{H} \otimes \mathcal{H})$ is entangled. Classically $\mathcal{U} \times \mathcal{V}$ exists canonically (Fubini); linearly, maximalization requires uncanonical choices on entangled data. We conjecture this failure of a linear Feferman–Vaught theorem is the structural reason no naive linear model theory exists, and note its family resemblance to the tensor-versus-commuting distinction underlying $\text{MIP}^* = \text{RE}$.

8. QUESTIONS

Question 8.1 (The quadrant question). Diagonalizable plufs have singleton S_π (Theorem 4.1); under CH some pluf has neither property (Theorem 5.9). Is there a *non-diagonalizable* pluf

with S_π a singleton? A positive answer produces a non-diagonalizable pure state whose lattice filter is maximal — strictly stronger than the known non-diagonalizable states of Akemann–Weaver [1] and Koszmider [12], none of which yield lattice filters at all (their states being singular, hence non-regular, the meet-stability of [5], Lemma 4.3 is unavailable). A negative answer makes RSP-for-all- T a *characterization* of diagonalizability: a lattice-level converse to Marcus–Spielman–Srivastava.

Question 8.2. CH enters Theorem 5.9 at exactly one point: countability of $\bigcup_{j < i} F_j$, which makes the h_n of Lemma 5.7 a decreasing ω -nest. Following the template of Farah–Wofsey [6], Theorem 6.46 — which derives a non-diagonalizable pure state from $\mathfrak{b} < \mathfrak{t}^*$, where $\mathfrak{t}^*$ is the least length of a maximal decreasing chain in the Calkin projection lattice — identify the pluf-analogue of $\mathfrak{t}^*$ (the least length of a maximal decreasing chain of ample subspaces, say) and derive Theorem 5.9 from an inequality against it, CH becoming a corollary. Is Theorem 5.9 a theorem of ZFC? Is it consistent that every pluf has RSP for every T ?

Question 8.3. Every nonprincipal pluf contains all finite-codimension subspaces (Lemma 2.2) and is invariant under finite-rank perturbations of its projections, but not obviously under compact ones — the gap being, once more, angle zero. Does the image of a nonprincipal pluf generate a maximal filter in the projection lattice of the Calkin algebra? Does every maximal filter there so arise? Either answer bears on Question 6.39 of [6] (whether every maximal Calkin filter lifts to a commuting family); in particular, is the image of the ample pluf of Theorem 5.9 maximal but not liftable?

Question 8.4. Kadison and Singer proved non-uniqueness of pure state extensions from the continuous masa $L^\infty[0, 1]$ [10]. Does some pluf extending the filter of spectral subspaces of the continuous masa have non-degenerate S_π in ZFC? (The countable chain $\{L^2[0, t_n]\}$ alone cannot witness this: any countable nested family is simultaneously coordinatizable, hence intimate for the coordinatizing basis [8]. The full family $\{L^2[0, t]\}_{t \in (0, 1]}$ generates the continuous masa and is not coordinatizable.)

Formal verification. This paper has been formalized and machine-checked in Lean 4 with Mathlib (toolchain v4.28.0), as part of a verification of this series; the artifacts compile with no unproved obligations, and every statement’s axiom footprint is exactly `propext`, `Classical.choice`, `Quot.sound`. In Sections 2–4: Lemma 2.1 together with its converse (a proper, upward closed, meet-closed family of closed subspaces satisfying the criterion is maximal); Lemma 2.2 in all three clauses; Proposition 2.3 in both of the forms proved above — for $\mathcal{H}$ by the triple of subspaces, and for every space of finite dimension at least two by the principality argument, so that no appeal to Kochen–Specker is needed at any dimension; Proposition 2.4 in all four clauses, over the topology generated by the sets $\widehat{M}$; Lemma 3.1 in the ellipsoid form displayed, together with the equivalence of that form of the round-slice property with its Rayleigh-quotient form; Proposition 3.2; Proposition 3.3 in full, including nonemptiness by Hahn–Banach, the sandwich characterization, the face property, weak* compactness by Banach–Alaoglu, and the equivalence of the round-slice property for all T with singleness of S_π ; and Theorem 4.1, parametrized by a paving hypothesis in the sense of Anderson — the Kadison–Singer theorem of [14] being the quarantined classical import — together with the elementary half of the converse recorded in Remark 4.2, namely that block-witnessed round slices for every operator yield paving.

Since Mathlib provides no state theory applicable to the bounded operators of a real Hilbert space, states are rendered directly as continuous linear functionals that are nonnegative on operators with nonnegative quadratic form and take the value 1 at the identity; all the properties used above (monotonicity, adjoint invariance, Cauchy–Schwarz, and the compression

identity at a projection of value 1) are derived from that definition. Section 5 has since been formalized as well, and with it every assertion of this paper. Mathlib has no essential spectrum, no Calkin algebra, no Fredholm theory and no Borel functional calculus; the first was defined directly by Weyl sequences, and the last was circumvented rather than built. In place of the spectral subspace $K_\lambda^\xi(V)$ of Lemma 5.5 the mechanized development uses the closed span of an orthonormal sequence of approximate eigenvectors with summable defects, which supplies exactly the three properties the argument consumes: infinite dimension, a bound $\|Tx - \lambda x\| \leq \delta\|x\|$ valid on the whole span, and infinite codimension of the intersection with N . Lemma 5.2 is verified in the ellipsoid form displayed here; Lemmas 5.3, 5.4 and 5.5 as stated; the blocking lemma, Lemma 5.7, in full, its recursion carried by a dependent-choice gadget whose stage predicate refers only to the vectors already chosen; and Theorem 5.9 and Proposition 5.11 by a transfinite recursion in which the three cases of the construction are internalized in a single stage lemma, so that the recursion itself never branches. The continuum hypothesis enters as an equinumerosity hypothesis on the statements that use it and is nowhere an axiom of the development. The mechanization also fixed two points of economy now incorporated above: Remark 5.8, that part (ii) of the blocking lemma consumes only the diagonal values; and the reduction of Proposition 5.11 to the ray $\mathbb{R}T$. Not formalized: §7, and the summary §6, whose contents are theorems of [9] and are verified there. The verification artifacts will be archived with the final version of this paper.

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